Capstone EDA — Grading Scoresheet

GA 5800 — Supervised-Learning-Readiness EDA

> **Purpose of this document.** This is a grader's scoresheet. The point structure mirrors the rubric embedded in the student-facing assignment, but this version expands each stage with four-level anchor descriptors, evidence checkpoints, and partial-credit examples. Use it for consistent scoring across submissions.

Student: _____ **Submitted on time?** ☐ Yes ☐ No (penalty: ____) **Total:** ___ **/ 100**

How to Use This Sheet

1. For each stage, check off the **Evidence Checkpoints** the student met.
2. Place the stage in one of the four anchor levels.
3. Multiply by the stage weight to assign points.
4. Apply automatic deductions at the bottom.
5. Apply lateness penalty last.
6. Record cross-cutting comments in the margin notes.

Anchor levels (consistent across all stages):

Level	% of stage points	Meaning
Exemplary	90–100%	All required elements; correct method-to-data matching; effect sizes; interpretation goes beyond description
Proficient	75–89%	All elements; minor methodological imperfection; interpretation correct but shallow in places
Developing	60–74%	Missing or wrong elements; statistics not matched to scale; effect sizes absent; interpretation = narration
Insufficient	< 60%	Major gaps; methodological errors that invalidate conclusions; or generic / not engaged with this dataset

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Stage 1 — Problem Framing & Target Definition *(/5)*

Evidence checkpoints:

- ☐ One-paragraph problem statement (audience + decision named)
- ☐ Target variable explicitly defined (name, type, prediction unit)
- ☐ Time-of-prediction stated (separates pre- vs. post-target features)
- ☐ Evaluation metric proposed **with justification**
- ☐ Single- vs. multi-target/output stated

Common partial-credit cases:

- Says "we will predict X" but never defines the time-of-prediction → cap at Proficient.
- Names a metric without justification → cap at Developing.
- Treats a regression target as classification or vice versa → Insufficient.

Score: ____ / 5 **Notes:** _____

Stage 2 — Data Inventory & Schema Audit *(/7)*

Evidence checkpoints:

- ☐ Shape, memory, dtype table
- ☐ Semantic feature-type table with `name | dtype | semantic type | role | notes`
- ☐ Semantic types include the full vocabulary (continuous, count, ordinal, nominal, binary, datetime, text, identifier, geographic)
- ☐ Disagreements between pandas dtype and semantic type are listed
- ☐ Coercion plan is stated for each disagreement

Common partial-credit cases:

- Treats every `object` column as "categorical" without distinguishing nominal vs. ordinal vs. text vs. ID → cap at Developing.
- Skips semantic-type column entirely → cap at Developing.
- Excellent table but no coercion plan → Proficient.

Score: ____ / 7 **Notes:** _____

Stage 3 — Data Quality Assessment *(/8)*

Evidence checkpoints:

- ☐ Exact-duplicate count
- ☐ Approximate / near-duplicate analysis with stated key
- ☐ Constant / quasi-constant columns identified
- ☐ ID-like columns identified
- ☐ Out-of-domain / impossible values flagged with severity
- ☐ Encoding glitches (whitespace, case, mojibake, mixed units) addressed
- ☐ Per-issue remediation proposed (drop / cap / impute / flag)

Common partial-credit cases:

- Reports `df.duplicated().sum()` only, no near-duplicate reasoning → cap at Proficient.
- Lists issues without proposing remediation → cap at Developing.
- No severity ranking → −1 within Proficient.

Score: ____ / 8 **Notes:** _____

Stage 4 — Missingness Analysis *(/8)*

Evidence checkpoints:

- ☐ Per-column missing % and row-completeness distribution
- ☐ Missingness matrix or correlation heatmap
- ☐ Missingness-vs-target test (chi-square or KW/MW depending on task) with **effect size**
- ☐ MCAR/MAR/MNAR posture argued (not just stated)
- ☐ Per-column imputation plan with justification

Common partial-credit cases:

- Reports missing % only → Developing.
- Has the test but only p-value, no effect size → Proficient (apply −2 deduction if applicable).
- Imputes everything with median "because it's robust" → cap at Proficient.
- No missingness-vs-target analysis at all → Developing or below.

Score: ____ / 8 **Notes:** _____

Stage 5 — Univariate Analysis *(/8)*

Evidence checkpoints (numeric):

- ☐ Summary statistics including skew & kurtosis
- ☐ ≥ 8 distribution plots (hist+KDE or ECDF)
- ☐ Transformation evaluated for skewed features with before/after plots

Evidence checkpoints (categorical):

- ☐ Cardinality table
- ☐ Frequency tables for low-cardinality
- ☐ Rare-level audit with stated threshold and grouping plan
- ☐ "Numeric-looking categoricals" called out (zip, IDs, codes)

Common partial-credit cases:

- Numeric covered well, categoricals skipped → cap at Proficient (−2 to −3).
- Plots without skew/kurtosis numbers → −1.
- Rare-level audit missing → −1 to −2.
- Transformation proposed but not visualized → −1.

Score: ____ / 8 **Notes:** _____

Stage 6 — Target Analysis *(/10)*

Classification checkpoints:

- ☐ Class proportions + bar chart
- ☐ Imbalance ratio computed
- ☐ Imbalance response recommended **and tied to Stage 1 metric**
- ☐ Stratification implications for Stage 10

Regression checkpoints:

- ☐ Distribution + summary stats + skew/kurtosis
- ☐ Outlier / zero-inflation check
- ☐ Transformation evaluation with before/after
- ☐ Recommendation on modeling scale

Both:

- ☐ Temporal target plot if time axis exists; drift commentary

Common partial-credit cases:

- Reports class imbalance but doesn't connect to metric choice → cap at Proficient.
- Heavy regression skew left untreated without justification → −2.
- Time axis present but ignored → −2 to −3.

Score: ____ / 10 **Notes:** _____

Stage 7 — Bivariate Signals vs. Target *(/12)*

Evidence checkpoints:

- ☐ Test selection matches each variable-type pair (continuous/ordinal/nominal/binary × class/reg)
- ☐ Effect sizes reported (Cramér's V, η^2, Cohen's d, point-biserial r, or equivalent) **for every test**
- ☐ Ranked feature table with `feature | test | statistic | p | effect size | interpretation`
- ☐ ≥ 6 illustrative plots, each with takeaway caption
- ☐ Multiple-comparison risk acknowledged

Common partial-credit cases:

- Pearson on everything → cap at Developing (apply −4 deduction).
- p-values without effect sizes → cap at Developing.
- Effect sizes present but no interpretation → cap at Proficient.
- No multiple-comparison comment → −1.
- "All $p < 0.05$ → all features matter" reasoning → cap at Proficient and flag in margin.

Score: ____ / 12 **Notes:** _____

Stage 8 — Multivariate Structure *(/12)*

Evidence checkpoints:

- ☐ Pearson **and** Spearman matrices; disagreements interpreted
- ☐ VIF (or condition number) computed; thresholds stated
- ☐ Pairplot / facet on top features colored by target
- ☐ Multivariate outlier method (Isolation Forest or Mahalanobis)
- ☐ Top 1% multivariate outliers investigated, not just counted
- ☐ (Bonus consideration) PCA scree + interpretation

Common partial-credit cases:

- One correlation matrix, no Spearman → −2 within Proficient.
- VIF computed but no action proposed for VIF > 10 features → −1 to −2.
- Outlier scores produced but not investigated → cap at Developing.

Score: ____ / 12 **Notes:** _____

Stage 9 — Mutual Information & Leakage Screen *(/12)*

MI checkpoints:

- ☐ Correct estimator (`mutual_info_classif` vs `mutual_info_regression`)
- ☐ `discrete_features` mask passed correctly
- ☐ Categoricals encoded before MI (or correctly flagged as discrete)
- ☐ Top-20 MI bar chart
- ☐ Disagreement vs. linear ranking discussed for ≥ 3 features

Leakage screen checkpoints (graded heavily):

- ☐ Every top-quartile feature evaluated against the four leakage questions
- ☐ Leakage-suspect features flagged in Data Card
- ☐ Justification for retain/exclude decisions

Sanity check:

- ☐ Tree-based importance probe with permutation importance on held-out split
- ☐ Caveat written: this is not model selection

Common partial-credit cases:

- Wrong MI estimator → cap at Developing (apply −5 deduction).
- MI computed but not compared to linear ranking → cap at Proficient.
- Leakage screen done generically (one paragraph, not per-feature) → cap at Developing.
- A clearly leaky feature retained without flag → apply −6 deduction.
- "Sanity check" presented as a model recommendation → −2 and margin note.

Score: ____ / 12 **Notes:** _____

Stage 10 — SL-Readiness Deliverables *(/15)*

10a — Data Card (5 pts):

- ☐ One row per feature
- ☐ All required columns present (semantic type, %miss, univariate notes, bivariate signal, MI, multicoll flag, leakage flag, treatment, encoding, scaling, keep/drop)
- ☐ Treatments are specific, not "TBD"
- ☐ Internally consistent with earlier stages

10b — Split recommendation (3 pts):

- ☐ Strategy chosen and named (random / stratified / grouped / time / nested)
- ☐ Justified against leakage between splits
- ☐ Group-leakage and temporal validity addressed where relevant
- ☐ Fractions specified

10c — Preprocessing pipeline sketch (3 pts):

- ☐ Maps every kept feature to a transformer
- ☐ Specific enough to implement in `ColumnTransformer`
- ☐ Distinguishes fit-on-train-only operations

10d — Top 10 + risks (4 pts):

- ☐ Ranked top-10 with cited evidence (Stage 7 + Stage 9)
- ☐ Risks list covers leakage, multicollinearity, missingness, drift, fairness/proxies

Common partial-credit cases:

- Data Card missing leakage column → cap at Proficient.
- Split strategy "we'll do 80/20" with no analysis → cap at Developing (apply −5 deduction).
- Top-10 list without evidence citations → −2.
- No risks list → −2.

Score: ____ / 15 **Notes:** _____

Cross-Cutting Criteria *(/3)*

Criterion	1 pt	Notes
Reproducibility & code quality	___	Seed, runs end-to-end, clean code
Statistical rigor	___	Tests match scale; effect sizes everywhere
Writing & visualization	___	Captions explain takeaways; labels complete

Subtotal: ___ / 3

Automatic Deductions

Trigger	Penalty	Applied?
Notebook does not execute top-to-bottom	−10	☐
No random seed	−2	☐
Plots missing titles/axis labels	−2 each, max −6	☐ ___
Pearson-only in Stage 7	−4	☐
Wrong MI estimator	−5	☐
Leakage feature retained without flag	−6	☐

Trigger	Penalty	Applied?
No split recommendation	−5	☐
Memo missing / < 2 pages	−5	☐
Data Card missing	−5	☐

Deduction subtotal: −____

Bonus *(max +5, cannot exceed 100)*

Bonus	Awarded
Fairness / proxy-variable analysis (+2)	☐
Drift / temporal-stability analysis (+2)	☐
Insightful interactive artifact (+1)	☐

Bonus subtotal: +____

Final Tally

Component	Points
Stage 1	___ / 5
Stage 2	___ / 7
Stage 3	___ / 8
Stage 4	___ / 8
Stage 5	___ / 8
Stage 6	___ / 10
Stage 7	___ / 12
Stage 8	___ / 12
Stage 9	___ / 12
Stage 10	___ / 15
Cross-cutting	___ / 3
Subtotal	___ / 100
Deductions	− ___
Bonus	+ ___
Lateness	− ___%

Component	Points
Final	**___ / 100**

Grader Comments

Strengths:

Areas for improvement:

Most important next step for this student:

Letter grade equivalent (course curve): ____

Grader signature / initials: ____ Date: ____